

Multi-Agent System: Theories and Applications

Lecture 11: Biological Oscillations

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Outline

- 1 Goodwin-type Model of Single Biological Oscillator
- 2 Synchronization of Linearly Coupled Goodwin-type Oscillators

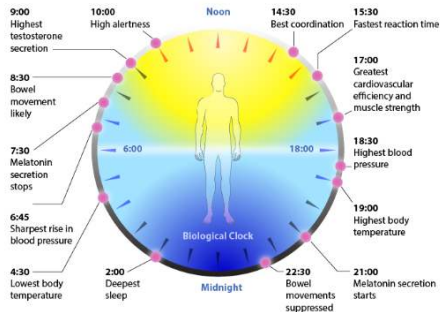
Goodwin-type Model

- q th-order model ($q \geq 3$):

$$\left\{ \begin{array}{l} \frac{dx_1}{dt} = f(x_q) - b_1 x_1 \\ \frac{dx_2}{dt} = x_1 - b_2 x_2 \\ \vdots \\ \frac{dx_q}{dt} = x_{q-1} - b_q x_q \end{array} \right.$$

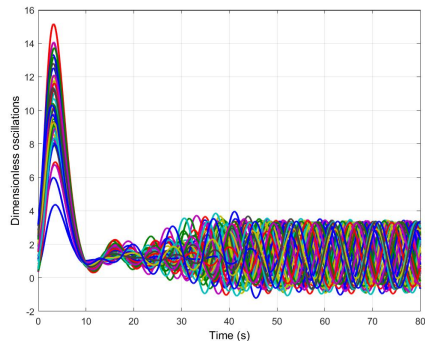
$$f(x_q) = \frac{1}{1 + x_q^p}, p > 0: \text{Hill function}$$

- Used in various biology systems, e.g. Testosterone hormone concentration, circadian rhythm, etc.



Biological oscillators without couplings are not synchronized

- Consider 50 Goodwin-type oscillators of 5th-order with $b_1 = b_2 = b_3 = b_4 = b_5 = 1, p = 10$.
- Initial conditions of oscillators are different → **unsynchronized**.
- Oscillations seem to be sinusoidal with intrinsic frequency 0.69.



Dinh Hoa Nguyen, Shinji Hara, "Entrainment analysis in Goodwin-type nonlinear oscillator networks driven by external periodic signals", *SICE Journal of Control, Measurement, and System Integration*, vol. 7, no. 6, pp. 337-346, 2014.

Linearly coupled model of Goodwin-type oscillators

- Model of each oscillator in Laplace domain:

$$z = h(s)u,$$

$$u = f(z),$$

where $h(s) = \frac{1}{(s+b_1)(s+b_2)\cdots(s+b_q)}$.

- With linear couplings between oscillators:

$$z = h(s)u,$$

$$y = f(z),$$

$$u = Ay,$$

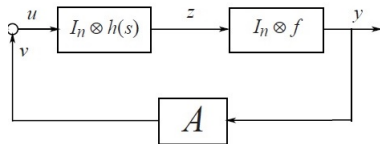


Diagram for linearly coupled Goodwin-type oscillator network

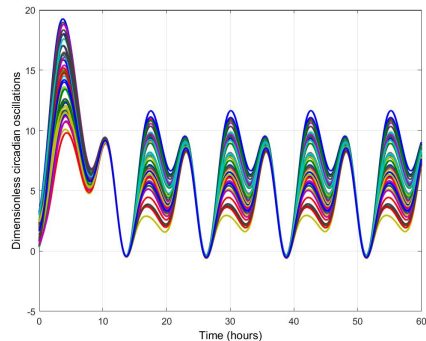
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Biological oscillators with couplings might not be synchronized

- Same Goodwin-type oscillators as in Slide 4.
- Couplings between oscillators are random → **unsynchronized**.
- **Oscillations are no longer sinusoidal.**

Questions:

- Q1: What are conditions for coupled Goodwin-type oscillators to be synchronized? (**intensively studied**)
- Q2: How coupled Goodwin-type oscillators can be synchronized? (**less investigated**) → **considered in this lecture**

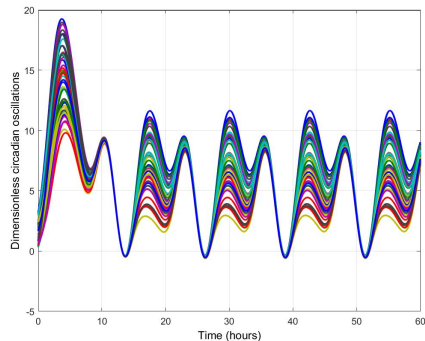


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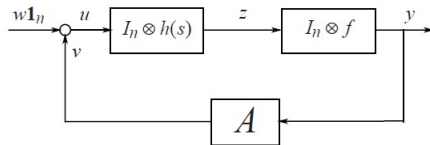
A hint for answering Q2 is from nature: biological clock for circadian rhythm inside human and other creatures is regulated by sunlight.

Entrainment of linearly coupled Goodwin-type oscillators

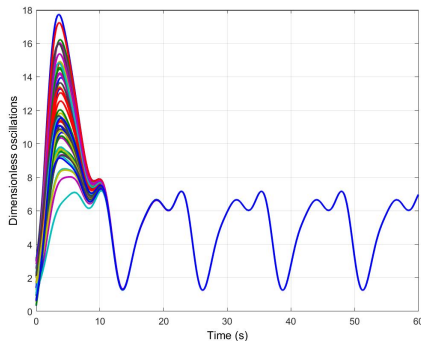
- Same Goodwin-type oscillators as in Slide 4.
- Coupling matrix between oscillators has an eigenvector $\mathbf{1}_{50}$.
- External input to each oscillator:
 $w(t) = 5 + 4 \sin(0.5t) + 7 \sin(t + \frac{4\pi}{11}) + 9 \sin(1.5t + \frac{6\pi}{7})$
 → synchronized.
- Oscillations are not sinusoidal.

Question:

- Q3: What is the synchronized oscillation profile?



Entrained Goodwin-type oscillator network



Entrainment analysis

Describing function (harmonic balance) method: approximate an oscillation by a sum of sinusoidal functions with high-order harmonic components.

- External input: $w(t) = \kappa_0 + \kappa_1 \sin(\omega t + \zeta_1) + \dots + \kappa_m \sin(m\omega t + \zeta_m)$
- For oscillator k ($k = 1, \dots, n$), approximate

$$z_k^{(m)}(t) \approx \alpha_{0k} + \alpha_{1k} \sin(\omega t + \varphi_{1k}) + \dots + \alpha_{mk} \sin(m\omega t + \varphi_{mk})$$

$$y_k^{(m)}(t) \approx \sigma_{0k} \alpha_{0k} + \sigma_{1k} \alpha_{1k} \sin(\omega t + \varphi_{1k}) + \dots + \sigma_{mk} \alpha_{mk} \sin(m\omega t + \varphi_{mk})$$

where σ_{0k} and $\sigma_{jk}, j = 1, \dots, m$ are describing functions calculated by

$$\sigma_{0k} = \frac{1}{2\pi\alpha_{0k}} \int_0^{2\pi} f(\alpha_{0k} + \dots + \alpha_{mk} \sin(mt)) dt,$$

$$\sigma_{jk} = \frac{1}{\pi\alpha_{jk}} \int_0^{2\pi} f(\alpha_{0k} + \dots + \alpha_{mk} \sin(mt)) \sin(jt) dt.$$

Entrainment analysis: Necessity

Denote

$$\left\{ \begin{array}{l} \alpha_0 = [\alpha_{01}, \dots, \alpha_{0n}]^T, \\ \alpha_j = [\alpha_{j1} e^{i\varphi_{j1}}, \dots, \alpha_{jn} e^{i\varphi_{jn}}]^T, j = 1, \dots, m, \\ \Sigma_0 = \text{diag}(\sigma_{0k})_{k=1, \dots, n}, \\ \Sigma_j = \text{diag}(\sigma_{jk})_{k=1, \dots, n}, j = 1, \dots, m, \\ \phi(s) \triangleq \frac{1}{h(s)}. \end{array} \right.$$

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Necessary Condition for Entrainment

If the entrainment occurs in the linearly coupled Goodwin-type oscillator network and the following harmonic balance equation is satisfied,

$$\left\{ \begin{array}{l} [\phi(0)I_n - A\Sigma_0] \alpha_0 = \kappa_0 \mathbf{1}_n, \\ [\phi(ij\omega)I_n - A\Sigma_j] \alpha_j = \kappa_j e^{i\zeta_j} \mathbf{1}_n, j = 1, \dots, m, \end{array} \right.$$

then the interconnection matrix A representing the couplings among Goodwin-type oscillators must have an eigenvector $\mathbf{1}_n$.

Dinh Hoa Nguyen, Shinji Hara, "Entrainment analysis in Goodwin-type nonlinear oscillator networks driven by external periodic signals", SICE Journal of Control, Measurement, and System Integration, vol. 7, no. 6, pp. 337-346, 2014.

Entrainment analysis: Sufficiency

Conjecture of Sufficient Condition for Entrainment

If the following conditions are satisfied,

- frequency of the external periodic input $w(t)$ is close to the natural frequency of network oscillations,
- amplitude of the external input is sufficiently large,
- interconnection matrix A has an eigenvector $\mathbf{1}_n$,

then the linearly coupled Goodwin-type oscillator network is entrained by the external input $w(t)$.

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Entrainment analysis – Case 1: A has an eigen-pair $(0, \mathbf{1}_n)$

- Denote $\rho_j = \hat{\varphi}_j - \zeta_j$ the phase shifts between the j th-order harmonics in the network oscillations and the external input, and $\phi_{R,j}, \phi_{I,j}$ the real and imaginary parts of $\phi(ij\omega)$, $i, j = 1, \dots, m$.
- Network dynamics becomes:

$$\begin{aligned}\phi(0)\hat{\alpha}_0 &= \kappa_0, \\ [\phi_{R,j} + i\phi_{I,j}]\hat{\alpha}_j e^{i\rho_j} &= \kappa_j, j = 1, \dots, m.\end{aligned}$$

Proposition

Amplitudes and phases of entrained oscillation can be estimated by:

$$\begin{aligned}\hat{\alpha}_j &= \frac{\kappa_j}{\sqrt{\phi_{R,j}^2 + \phi_{I,j}^2}} \Rightarrow \hat{\alpha}_j \text{ is a monotonically increasing function of } \kappa_j, \\ \hat{\varphi}_j &= \zeta_j - \text{atan}(\phi_{I,j}/\phi_{R,j}) \Rightarrow \rho_j \text{ is constant w.r.t } \kappa_j.\end{aligned}$$

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Entrainment analysis – Case 2: A has an eigen-pair $(\lambda, \mathbf{1}_n)$, $\lambda \neq 0$

No analytical results!

Entrainment analysis – Case 2: A has an eigen-pair $(\lambda, \mathbf{1}_n)$, $\lambda \neq 0$

No analytical results!

- Verifying the monotonic dependence of the entrained oscillations to the external input as it is varying.
- Sinusoidal external inputs, i.e., $w(t) = \kappa_0 + \kappa_1 \sin(\omega t + \zeta_1)$.
- Network dynamics:

$$\begin{aligned} \left[(\phi_{R,1} - \lambda \hat{\sigma}_1)^2 + \phi_{I,1}^2 \right] \hat{\alpha}_1^2 &= \kappa_1^2, \\ -\frac{\phi_{I,1}}{\phi_{R,1} - \lambda \hat{\sigma}_1} &= \tan(\rho_1). \end{aligned}$$

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Entrainment analysis – Case 2: A has an eigen-pair $(\lambda, \mathbf{1}_n)$, $\lambda \neq 0$ (Cont.)

- **Assumption:** The integral in the describing function $\hat{\sigma}_1$ is a monotonically decreasing function of $\hat{\alpha}_0$ and $\hat{\alpha}_1$.

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Proposition

Amplitude $\hat{\alpha}_1$ is a monotonically increasing function of κ_1 , if $\phi_{R,1}\lambda > 0$.

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Proposition

The phase shift ρ_1 is a monotonic function of the amplitude κ_1 as in the following table.

λ	$\phi_{R,1}$	$\phi_{I,1}$	ρ_1
–	–	–	– ↑
–	–	+	+ ↓
–	+	–	+ ↑
–	+	+	– ↓
+	–	–	– ↓
+	–	+	+ ↑
+	+	–	+ ↓
+	+	+	– ↑

Entrainment analysis for circadian network: Laplacian case

- 3rd-order Goodwin-type model for describing circadian network.
- $n = 3$, i.e. 3 oscillators.
- A is a Laplacian matrix, i.e. A has an eigen-pair $(0, \mathbf{1}_3)$.

Proposition

The phase shift ρ_1 is a monotonically decreasing function of the frequency ω of the entrainment signal (e.g. sunlight).

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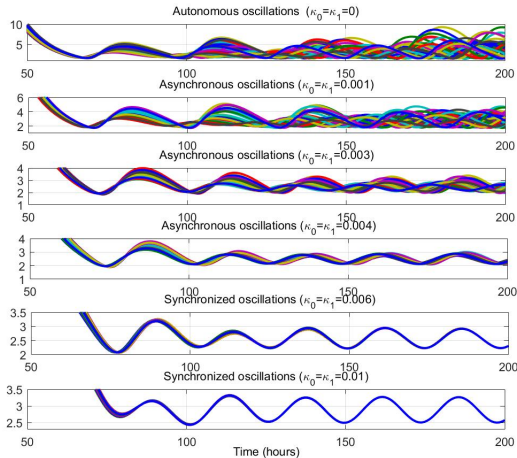
✧ This result agrees with realistic biological experiments in the literature.

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Entrainment analysis for circadian network: Laplacian case – Example

- Parameters for the model of circadian oscillations in *Neurospora crassa*:
 $b_1 = 0.2, b_2 = 0.2, b_3 = 0.1$.
- 50 oscillators coupled by $A = I_5 \otimes A_1 + L \otimes I_{10}$, L Laplacian, A_1 has eigenvector $\mathbf{1}_{50}$.

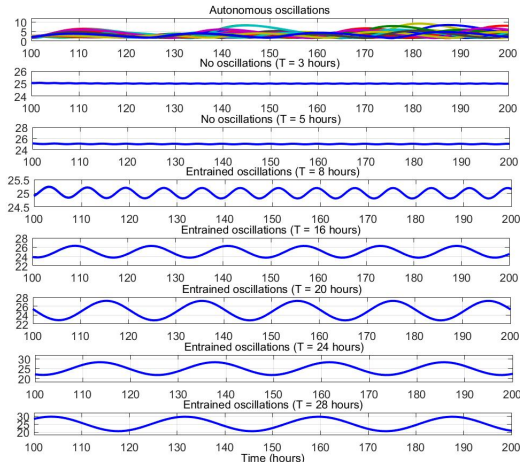
Entrainment as *amplitude* of external signal varies



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Entrainment as *frequency* of external signal varies



Entrainment analysis for circadian network: general case

- 3rd-order Goodwin-type model for describing circadian network.
- $n = 3$, i.e. 3 oscillators.
- A is not necessarily a Laplacian matrix.

Corollary

Amplitude $\hat{\alpha}_1$ is a monotonically increasing function of κ_1 , if $[b_1 b_2 b_3 - (b_1 + b_2 + b_3)\omega^2]\lambda > 0$.

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λ	ϕ_R	ϕ_I	$\hat{\phi}$
—	—	—	— $\uparrow$
—	—	+	— $\downarrow$
—	+	+	— $\downarrow$
+	—	—	— $\downarrow$
+	—	+	— $\uparrow$
+	+	+	— $\uparrow$

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- Parameters for the model of circadian oscillations in *Neurospora crassa*:
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- 50 oscillators coupled by $A = I_5 \otimes A_1 + L \otimes I_{10}$, L and A_1 are Laplacian.

Monotonicity of the phase shift of entrained circadian oscillations as *frequency* of external signal varies

